

Every voice agent picks one silence threshold.

Nobody published what it actually does.

Nod measures AssemblyAI's turn-detection knobs against the live service, and moves them mid-call. This deck is what the measurement found — including the part where it contradicted our own headline.

One threshold, tuned for an average speaker, applied to everyone who calls.

It cuts off the people who pause mid-sentence: older callers, non-native speakers, anyone reading a number off a card.

ladder_say_500.wav - hold 532 ms after "...my appointment to" - red = turn ended inside the pause - pilot_ladder.say.live.json

aggressive



2 turns

balanced



1 turn

conservative



1 turn

Same sentence. Same **532 ms** pause after "...my appointment to".

Three configs. The first splits it in **two**.

A benchmark that cannot run before the prediction is written down.

predict → commit → run

The runner exits 2 unless the predictions file already exists. A prediction made after the observation is not a prediction, so the ordering is a refusal in code rather than a habit — enforced at runtime, and for the first ladder visible in the history too (f3e7838 predicts, b7c3ccc observes).

One clock, on purpose

Firing times are measured from the acoustic end of the audio we are about to feed, not from the service's word timings — which is how the fourth fact on the next slide was found.

A fake for CI, a socket for numbers

Tests never touch the network (INV-7). Every published figure comes from the real API, labelled `.live.` in the filename so a chart cannot lose its provenance.

Mutation-tested where it lies

Coverage is deliberately uneven. Full discipline on anything feeding a published number; thinner on plumbing that fails visibly. A wrong number does not fail visibly — it prints a plausible one.

Four facts about the service.

The confidence threshold is inert for endpointing.

Arms at 0.0 and 1.0 landed 6 ms apart, the high arm *earlier*.

69 sessions

ADR-001

Turns end on a clamp, not on a gate.

Two parameters, thirteen live rows, all inside 120 ms.

16 sessions

ADR-055

min_turn_silence is continuous to at least 2.4 s.

777 → 2574 ms of patience. max_turn_silence only binds below C, or at end of stream.

4 sessions

ADR-055

The word-timing field is not a silence anchor.

Prefix end reported **560 ms late**, continuation **190–250 ms early** — same session, so not an offset.

16 sessions

ADR-055

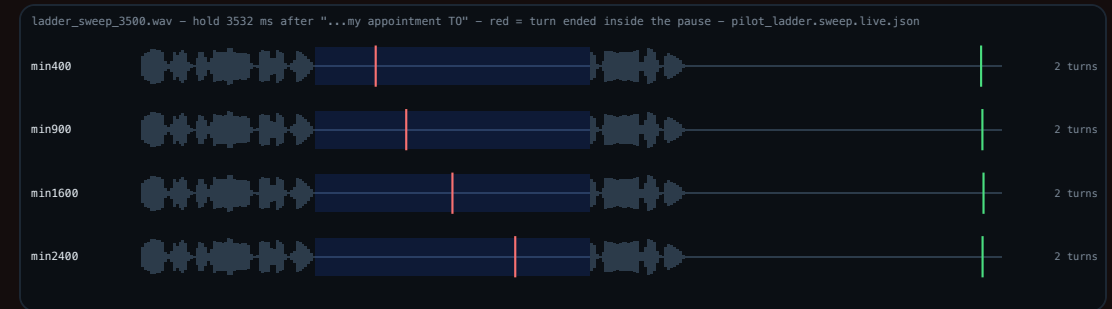
```
fire_at = prefix_end + clamp(C, min_turn_silence, max_turn_silence) + overhead
C ≈ 590 ms · overhead ≈ 175 ms
```

Last week the headline was ~~"the regime is unreachable".~~

ADR-054. Twelve sessions. Firing time did not move with the pause, so `max_turn_silence` never engages, so *"Nod may be a correct controller for a gate that does not engage."*

ADR-055. The observation reproduced exactly. The conclusion was wrong. **Two knobs bracket the same quantity and we varied one.** Sweeping the other took four sessions and six minutes.

The correction also says **our own controller points at the wrong knob**, and `MIN_MS_CEIL = 900` caps the working one at a third of its range. Both are still in the code — retuning a control law on one clip of synthetic speech is tuning by ear.



The red line is where the turn ended. The blue band is the caller's pause. Only `min_turn_silence` changes.

Built to exercise the instrument.

A two-gate control law, a per-caller rhythm profiler on fixed-capacity ring buffers, and a session proxy that rewrites turn-detection config mid-stream — out of band, never in the audio path.

What is built and tested

Bounded `decide()`, p99 under 5 ms, no I/O and no LLM in the decision path. Constant memory per session. Every config change carries a reason record. Fails soft in a call, loud in dev.

What has not been shown

That the closed loop helps a real caller. The controller has never run on human speech, and on the one live sweep the controlled arms were identical to the baseline — the corpus cannot warm the profiler, and the patch census is **unmeasured, not zero**.

What this is not.

- **The corpus is one macOS say voice**, and its own manifest says it is *"not adequate for ... any published number"*. Every figure here comes from it anyway, and that is stated rather than discovered.
- **Nod has never been run on human speech**. Track C is written and unrecorded; Track B, the real atypical-speech corpus, is cut. The project is about callers it has not met.
- **Our controller is aimed at the wrong knob** and caps the right one at 900 ms against a demonstrated 2.4 s. Found by measurement, not fixed.
- **Live PCR attributes boundaries to the wrong gap**, because it reads the word-timing field in fact 4. TTL and FRAG do not use it and are unaffected.
- **740 tests, 98.18 % coverage, 182/182 mutations killed** certify less than they look: lines *executed*, not behaviour asserted, and mutation coverage is file-granular — a kill proves *some* test noticed, never which.
- **Nod is not the first adaptive endpointer**. AssemblyAI's own Voice Agent API already ships adaptive pacing. Nod is an open one on the streaming STT path, with the policy written down and the numbers published.